Replicating DreamBooth: Subject-Driven Fine-Tuning of Stable Diffusion

Ryan Qiu Gordon Mei Caleb Shim Evan Cui
Cornell University CS4782 — Deep Learning

https://github.com/caleb-bit/dreambooth-replication

Introduction/Motivation

Large-scale text-to-image diffusion models synthesize photorealistic images from prompts but cannot render a *specific* subject unseen during training. Prompting "a photo of my dog in Paris" produces a plausible dog, but not the specific dog that the user wants. Even exhaustive description (breed, color, distinguishing markings) cannot uniquely identify a specific instance, since the model has no reference to bind to. **DreamBooth** [1] (Ruiz et al., CVPR 2023) addresses this by fine-tuning the model on 3–5 reference images, binding a particular subject to a rare identifier token and enabling compositional generation in novel contexts.

We replicate DreamBooth on Stable Diffusion v1.5 (SD 1.5) [2], implementing the full training pipeline from scratch. Our goal is to reproduce the paper's reported quantitative metrics on SD 1.5 and document the implementation details necessary to achieve them.

Chosen Result

We target the quantitative evaluation from Table 2 of the DreamBooth paper, which reports subject fidelity (DINO [4], CLIP-I [3]) and prompt fidelity (CLIP-T) for fine-tuned SD 1.5 models across 30 subjects. The paper reports mean scores of DINO 0.668, CLIP-I 0.803, and CLIP-T 0.305 for SD. These results show that a fine-tuned model can preserve specific subject identity while remaining responsive to novel prompts. Reproducing these scores provides evidence that our pipeline correctly implements key training components, such as rare-token binding and prior preservation mechanisms.

Methodology

Architecture. SD 1.5 has three components: a VAE (≈83M params) that encodes images to latents, a UNet (≈860M params) that denoises in latent space, and a CLIP text encoder (≈123M params). We fine-tune the UNet only; both the VAE and CLIP text encoder are frozen throughout training.

Training objective. Instance images are captioned as "a photo of `sks` [class]", where `sks` is a rare identifier with negligible prior semantics. The model has no strong associations with this token, so it can be used to represent the new subject. The class label (e.g. "dog") anchors `sks` within the correct semantic area, ensuring the model understands the general category while learning to associate the rare token with the specific instance. At each step, both instance and class images are encoded to latents, noise is sampled and added, and the UNet predicts the noise. The combined loss is:

$$\mathcal{L} = \underbrace{\|\hat{\varepsilon}_{\text{inst}} - \varepsilon_{\text{inst}}\|^2}_{\text{instance loss}} + \lambda \underbrace{\|\hat{\varepsilon}_{\text{class}} - \varepsilon_{\text{class}}\|^2}_{\text{prior preservation}}, \quad \lambda = 1.0.$$

Prior-preservation class images (100 per subject) are generated by the unmodified SD 1.5 model before training, substituting for the Google Imagen model used in the original paper.

Hyperparameters. 800 steps, learning rate 5×10^{-6}, resolution 512×512, batch size 1, 8-bit AdamW optimizer [5], gradient checkpointing. The UNet runs in FP32 while frozen components use FP16.

Dataset and evaluation. We fine-tune on `dog`, `dog2`, and `backpack` from the official DreamBooth dataset [7], each with 5 instance images. Evaluation generates 4 images per prompt across 5 prompts (20 images/subject). DINO uses ViT-S/16 [4]; CLIP-I and CLIP-T use CLIP ViT-L/14 [3]. Training runs on a Google Colab T4 GPU (≈35 min/subject); metrics are computed on CPU.

Results & Analysis

Our mean DINO (0.598) falls slightly below the paper's 0.668, which is expected given that we evaluate only 3 subjects rather than the full 30. CLIP-I (0.861) exceeds the paper's 0.803, indicating strong

Table 1: Quantitative metrics. "Real images" is the upper bound; "Paper (SD)" is the DreamBooth paper's SD 1.5 result.

	DINO↑	**CLIP-I**↑	**CLIP-T**↑
dog	0.716	0.867	0.275
dog2	0.542	0.828	0.247
backpack	0.536	0.887	0.274
Ours (mean)	**0.598**	**0.861**	**0.265**
Real images	0.774	0.885	—
Paper (SD)	0.668	0.803	0.305

semantic alignment between generated and reference images. CLIP-T (0.265) is below the paper's 0.305, reflecting the inherent subject–prompt fidelity trade-off: our fine-tuned models preserve identity well but sacrifice some text adherence. Qualitatively, generated images maintain subject appearance across novel backgrounds and lighting; see Figure 1.

The most notable gap is DINO variance across subjects: `dog` scores 0.716 (near the paper's mean) while `dog2` and `backpack` score around 0.54, suggesting that the model's ability to maintain feature identity is highly dependent on the reference images. The backpack scores high on CLIP-I (0.887) despite lower DINO, indicating that semantic similarity is preserved even where exact feature identity is weaker.

Overall, this replication demonstrated that we are able to achieve significant evaluation thresholds for subject-driven generation even when training on a smaller diffusion model like SD 1.5 on a smaller set of generated samples.

Reflections

Prior preservation. Prior loss is a critical part of the loss function that ensures that the model maintains its class representations, as observed by the ablation.

FP16 causes silent gradient underflow. Using half-precision for the UNet caused gradient updates to fall below the representable range, halting weight updates with no error or warning. Keeping the UNet in FP32 resolved this.

Freezing the text encoder. Jointly fine-tuning the UNet and CLIP text encoder led to overfitting: outputs lost compositional flexibility and ignored novel prompts. The text encoder already encodes language structure; updating it causes the model to redefine token semantics rather than adjust the visual pathway.

Logo fine-tuning. We applied DreamBooth to a custom `CS4782` course logo using 5 augmented variants as instance images, with `sks` as the identifier and `logo` as the class noun. Direct generation ("a photo of a `sks` logo") reproduced the graphic faithfully across samples, with only minor color variation. Context placement failed: prompts such as "a man wearing a `sks` logo" or "a mug with a `sks` logo" caused the model to render `sks` as literal glyphs rather than the learned graphic. It may be useful to include contextual mockups in the reference set (e.g., the logo composited onto a shirt, paired with a matching prompt), so the model sees the subject in context during training rather than only as an isolated graphic.

Future work. Scaling to 30 subjects enables direct benchmark comparison; ablations over step count and learning rate would map the underfitting/overfitting boundary.

References

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(a) Reference subject

(b) “a photo of [v] dog with Cornell merch”

Figure 1: Reference image and a generated image using the fine-tuned model, demonstrating subject identity preserved in a novel context.